

Zoe Ko

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Education

- Johns Hopkins University** 2024 – Present
Physics and Astronomy Ph.D. Student (current)
M.A. Physics (2026)
- University of California, Berkeley** 2019 – 2023
B.A. Astrophysics
B.A. Data Science
High Distinction in General Scholarship (magna cum laude equivalent)

Research Experience

- Lawrence Livermore National Lab, NASA Pandora Team** March 2026 – Present
Graduate Intern | Advisor: Peter McGill
 - Develop a Gaussian Process framework to extract spot and photosphere spectra from the NASA Pandora mission to characterize host stars and improve exoplanetary atmospheric characterization.
- Johns Hopkins University** August 2024 – Present
Research Assistant | Advisor: Néstor Espinoza
 - Implemented a Gaussian Process-aided atmospheric retrieval framework to mitigate stellar contamination in transmission spectroscopy.
 - Applied retrieval framework to constrain the atmospheric properties of a gas giant orbiting an M dwarf to better understand giant planet formation around low mass stars.
- California Institute of Technology LIGO Lab** June 2022 – January 2024
Research Intern | Advisor: Katerina Chatziioannou
 - Developed and implemented a hierarchical Bayesian inference framework to analyze binary black hole spin distributions using LIGO gravitational wave data.
 - Generated mock binary black hole populations with different component spin parameters to quantify how much spin information is extractable from observed effective spins.
- University of California, Berkeley SETI Research Center** June 2021 – January 2023
Research Intern | Advisor: Howard Isaacson
 - Developed a laser detection algorithm to search for potential technosignatures in optical stellar spectra and applied algorithm to over 1000 spectra of nearby stars to set detection limits.
- Pennsylvania State University** September 2019 – February 2023
Research Assistant | Advisor: Ian Czekala
 - Designed a pipeline from scratch to detect radial velocity variations through cross-correlation techniques and applied pipeline to 28 candidate binary star systems.

Awards and Fellowships

- William H. Miller III Graduate Fellowship** August 2024 – August 2025
 - Awarded a 12-month fellowship from Johns Hopkins to support independent graduate research.
- Fulbright Scholarship Award** September 2023 – July 2024
 - Designed and taught an English language curriculum for seventh and tenth grade students in Thailand.

Publications

- Z. Ko**, N. Espinoza, A. Jordán, E. M. Bryant, M. Schlecker, M. J. Hobson, R. Brahm, R. Burn, & T. Henning. 2026, "JWST Transmission Spectroscopy of TOI-3235 b: Constraints on Giant Planet Formation around M Dwarfs amid Stellar Contamination," AJ, under review.
- S. J. Miller, **Z. Ko**, T. Callister, & K. Chatziioannou. 2024, "Gravitational waves carry information beyond

effective spin parameters but it is hard to extract,” Phys. Rev. D, 109, 104036

- A. Zuckerman, **Z. Ko**, H. Isaacson, S. Croft, D. Price, M. Lebofsky, & A. Siemion. 2023, “The Breakthrough Listen Search for Intelligent Life: A Laser Search Pipeline for the Automated Planet Finder,” AJ, 165, 114

Conference Talks and Posters

- **Z. Ko**, N. Espinoza, A. Jordan, P. McGill. JWST Transmission Spectrum of a GEMS Target, Exoplanets 6, Porto, Portugal, June 2026. (Poster)
- **Z. Ko**, P. McGill. Mitigating Stellar Contamination with Pandora Data, Pandora Team Meeting, NASA Goddard, Greenbelt, MD, June 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Probing the Atmospheres of Giant Planets around M Stars Amid Stellar Contamination, Invited Speaker, Universidad de Chile, Santiago, Chile, March 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Characterizing Giant Planets around M Dwarfs amid Stellar Contamination with Gaussian Processes, Invited Speaker, Pontifical Catholic University, Santiago, Chile, March 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Unveiling the Atmospheres of Giant Planets around M Stars Amid Stellar Contamination, Invited Speaker, Universidad Diego Portales, Santiago, Chile, March 2026. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. Mitigating Stellar Contamination in JWST Transit Data with Gaussian Processes, Exojamboree, Baltimore, MD, November 2025. (Oral Presentation)
- **Z. Ko**, N. Espinoza, A. Jordan. A Gaussian Process Framework for Exoplanet Atmospheric Retrievals, Exoclines VII, Montreal, QC, Canada, July 2025. (Poster)
- **Z. Ko**, N. Espinoza, A. Jordan. Gaussian Process Retrievals for TOI-3235b, Emerging Researchers in Exoplanet Science Symposium X, Princeton University, Princeton, NJ, June 2025. (Oral Presentation)
- **Z. Ko**, S. J. Miller, K. Chatziioannou. Analyzing the Effective and Component Spin Distributions of Binary Black Hole Mergers, American Physical Society April Meeting, Minneapolis, MN, April 2023. (Oral Presentation)
- **Z. Ko**, H. Isaacson, A. Zuckerman, S. Croft. Search for Laser Emission Lines with the Automated Planet Finder Telescope, American Astronomical Society Meeting, Seattle, WA, January 2023. (Poster)
- **Z. Ko**, H. Isaacson, A. Zuckerman. Search for Laser Emission Lines with the Automated Planet Finder Telescope, AbSciCon, Atlanta, GA, May 2022. (Poster)
- **Z. Ko**, H. Isaacson, A. Zuckerman, S. Croft. Search for Laser Emission Lines with the APF, Order of the Octopus, Virtual, August 2021. (Poster)

Skills

Programming: Advanced proficiency in Python (NumPy, SciPy, pandas, matplotlib, Astropy, etc.), Java, R.

Statistical Modeling: Extensive experience with Bayesian inference frameworks (PyMC, emcee, dynesty) and Gaussian process modeling (tinygp, george, celerite).

Languages: Native fluency in English and Mandarin; strong proficiency in Spanish and Thai.

Teaching and Outreach

STEM Mentor at learnSTEM Research Mentorship Program March 2026 – June 2026

- Mentor a high school student in a 9-week program to complete an independent research project.

Physics and Astronomy Peer Mentor at Johns Hopkins August 2025 – May 2026

- Advise physics and astronomy undergraduate students academically and professionally.

Peer Tutor at the Berkeley Student Learning Center August 2022 – May 2023

- Provided weekly tutoring services for introductory astrophysics courses.

Research Mentor at the Berkeley Undergraduate Lab August 2021 – May 2022

- Mentored five undergraduates through a year-long independent research project on binary star systems.

Student Mentor and Outreach Director of Berkeley Engineers and Mentors January 2020 – May 2023

- Led weekly science labs at elementary and middle schools in Oakland and coordinated outreach events.